NATIONAL UNIVERSITY OF
COMPUTER & EMERGING SCIENCE

CARPER-THE CAR PROBLEM SOLVER

CARPER TEAM

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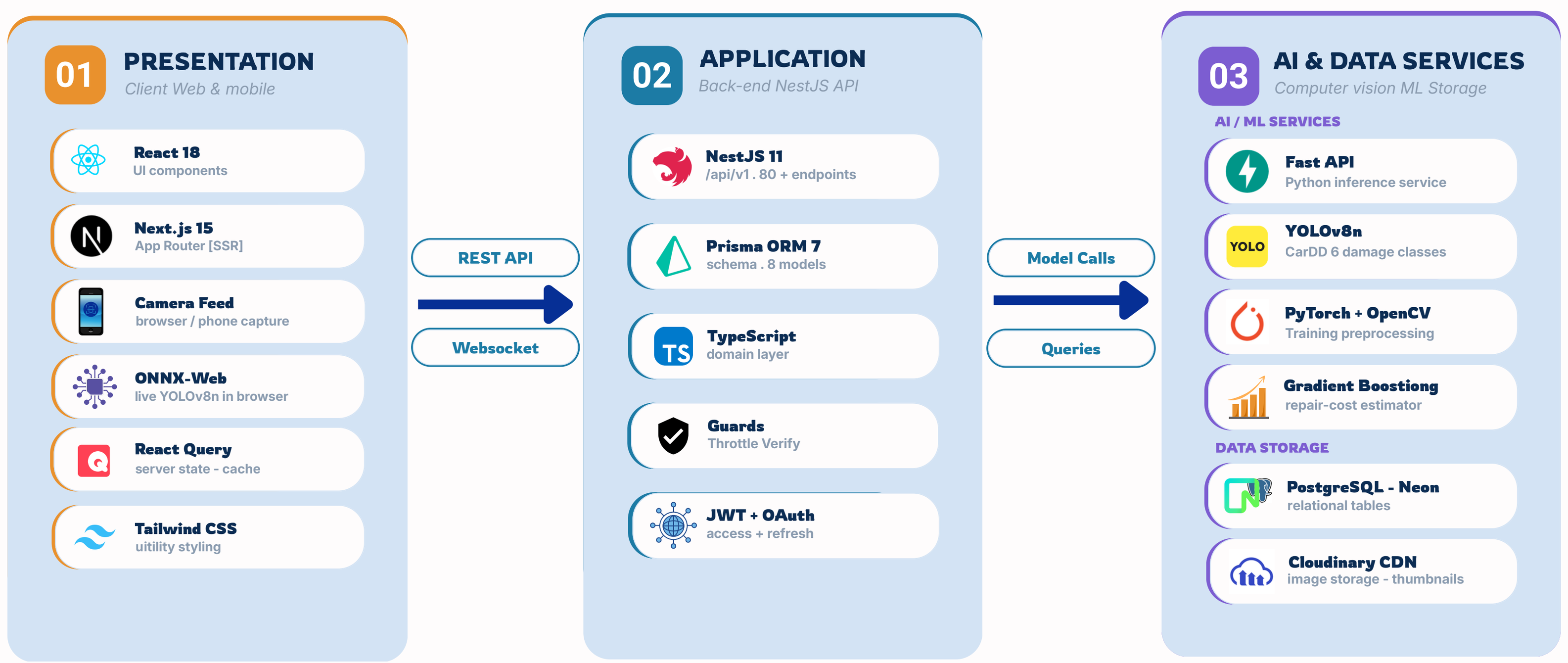
ABSTRACT

With the rapid advancement in artificial intelligence and computer vision, automated vehicle damage detection has emerged as a significant area of research and application. Manual car damage assessment has limitations such as, it is prone to human-error, fraud, and is time-consuming. These issues affect industries such as insurance, vehicle rental and resale. CARPER– The Car Problem Solver is a smart web-based application that uses computer vision and data driven techniques to address these issues by detecting exterior vehicle damage and gives an estimated repair cost. The application uses YOLOv8n object detection model trained on CarDD dataset, which comprises 4,000 high resolution images with following six damage categories: dent, scratch, crack, glass shatter, tire flat and lamp broken. For cost estimation multiple regression models were evaluated and on the basis of the best performance Gradient Boosting was used. Semantic search and price-matching algorithms connect users to affordable replacement parts from online marketplaces. The system was evaluated mainly on mean Average Precision (mAP) metrics. CARPER delivers a comprehensive , transparent and accessible solution for automobile owners by automating damage detection and cost estimation.

SYSTEM ARCHITECTURE



OBJECTIVE

The Project aims to web-based automated vehicle damage inspection by:

- Detecting six types of external car damage using YOLOv8n on CarDD
  (Dent, Scratch, Crack, Glass Shatter, Tire Flat & Lamp Broken)
- Estimating repair costs via Gradient Boosting
- Recommending affordable parts via semantic marketplace search
- Providing a seamless web interface for car owners.

RESULTS

1.Yolov8n

The YOLOv8 model was initially trained for 100 epochs using an image size of 640 × 640 and a batch size of 16. A fine-tuning phase was then performed for 50 additional epochs using a higher image resolution of 832 × 832 to improve detection of small damage regions such as cracks and scratches. During fine-tuning, the batch size was reduced to 8 and the learning rate was lowered to 0.001 for more stable optimization. Following are the results:

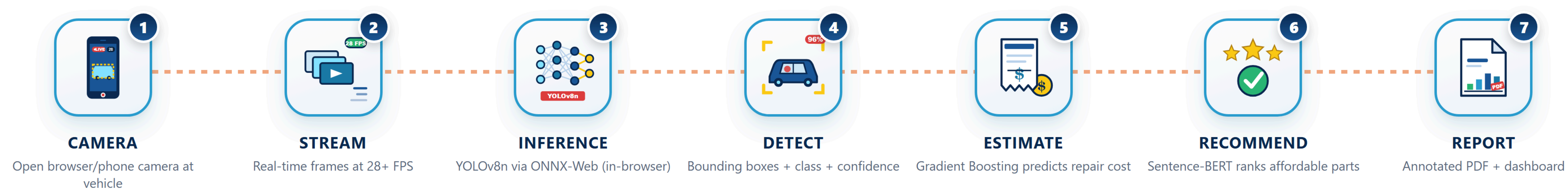
Class	Images	Instances	Precision	Recall	mAP50	mAP50:95
All	374	785	74.3	70.4	73.2	56.1
Dent	157	236	63.2	57.2	58.1	33.2
Scratch	183	307	57.7	57.9	54.5	29.1
Crack	48	70	50.8	47.1	48.3	23.6
Glass Shatter	71	71	93.2	95.8	97.9	88.1
Lamp Broken	65	69	91.6	76.8	87.1	72.6
Tire Flat	31	32	89.4	87.5	93.4	89.7

2.Regression Models

Multiple regression models were evaluated for vehicle repair cost prediction, including Linear Regression, Random Forest, Extra Trees, and Gradient Boosting. Model performance was assessed using MAE, RMSE, and R² evaluation metrics. Among all models, Gradient Boosting achieved the best overall performance with the highest R² score and lowest prediction error. Cross-validation results also demonstrated strong model stability and generalization capability.

Regression Model	MAE (PKR)	RMSE (PKR)	R² SCORE
Linear Regression	4,049	5,496	96.5
Random Forest	4,145	5,951	95.9
Extra Trees	3,874	5,150	96.9
Gradient Boosting	3,523	4,802	97.3

WORKING PIPELINE



WORKING MEHODOLOGY

The user registers, verifies their email, and logs into CARPER, then navigates to the live detection page where YOLOv8n runs directly in the browser via ONNX Web, detecting vehicle damage and overlaying bounding boxes in real time. Detected damage classes are passed to the Gradient Boosting model along with car information to generate a repair cost estimate in PKR, while a semantic search module simultaneously finds affordable replacement parts. All results are compiled into a downloadable PDF report and interactive dashboard.

CONCLUSION

CARPER successfully demonstrates that a lightweight YOLOv8n model trained on the CarDD dataset can deliver competitive multi-class vehicle damage detection at real-time inference, achieving an overall mAP@50 of 73.2 with with strong performance on visually distinct damage categories, Glass Shatter (97.9), Tire Flat (93.4), Lamp Broken (87.1), Dent (58.1), Scratch (54.5) & Crack (48.3). Inference was measured at 2.1 ms preprocessing, 4.4 ms inference, and 1.5 ms postprocessing per image The integrated cost estimation module and semantic part-recommendation pipeline further extend CARPER into a practical, end-to-end automotive assessment tool. The system fulfils all three project success criteria and is presented as a fully containerised, web-deployed solution suitable for insurance automation, vehicle rental services, and private resale valuation.

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